School of Computing
National University of Singapore
CS5340: Uncertainty Modeling in AI
Semester 1, AY 2025/26

Exercise 2

Question 1 (Lecture 3 and 6)

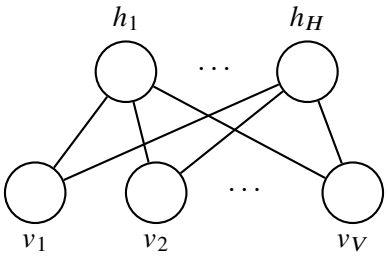


Figure 1.1

The restricted Boltzmann machine is a Markov Random Field (MRF) defined on a bipartite graph as shown in Figure 1.1. It consists of a layer of visible variables $\mathbf{v} = [v_1, \ldots, v_V]^T$ and hidden variables $\mathbf{h} = [h_1, \ldots, h_H]^T$, where all variables are binary taking states $\{0, 1\}$. The joint distribution of the MRF is given by:

$$p(\mathbf{v}, \mathbf{h}) = \frac{1}{Z(\mathbf{W}, \mathbf{a}, \mathbf{b})} \exp\left(\mathbf{v}^T \mathbf{W} \mathbf{h} + \mathbf{a}^T \mathbf{v} + \mathbf{b}^T \mathbf{h}\right),$$

where $\theta = \{\mathbf{W}_{V \times H}, \mathbf{a}_{V \times 1}, \mathbf{b}_{H \times 1}\}$ are the parameters of the potential functions, and $Z(\cdot)$ is the partition function.

a) Given that:

$$p(h_i = 1 \mid \mathbf{v}) = \sigma\left(b_i + \sum_j W_{ji} v_j\right),$$

where $\sigma(x) = \dfrac{e^x}{1 + e^x}$ is the sigmoid activation function. Show that the distribution of hidden units conditioned on the visible units factorizes as:

$$p(\mathbf{h} \mid \mathbf{v}) = \prod_i p(h_i \mid \mathbf{v}).$$

Show all your workings clearly.

b) Assuming that the restricted Boltzmann machine consists of only 2 visible and 1 hidden variables, and the joint distribution of the MRF is given by:

h	v_1	v_2	$\exp\left(\mathbf{v}^T \mathbf{W} h + \mathbf{a}^T \mathbf{v} + bh\right)$
0	0	0	1.00
0	0	1	2.13
0	1	0	4.65
0	1	1	9.90
1	0	0	3.65
1	0	1	8.66
1	1	0	4.22
1	1	1	10.01

Find the unknown parameters, i.e. $\theta = \{\mathbf{W}_{2\times1}, \mathbf{a}_{2\times1}, b\}$.

Question 2 (Lecture 8)

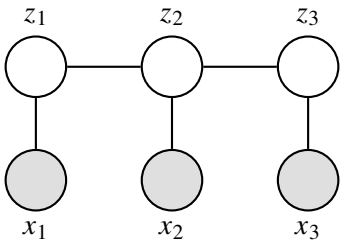


Figure 2.1

Figure 2.1 shows a Markov Random Field (MRF) representation of a Hidden Markov Model (HMM) over three time steps. The hidden variables z_1, z_2, z_3 are discrete random variables that take three possible states $z_n \in \{F, H, M\}$, and x_1, x_2, x_3 are the observed variables that take on real values $x_n \in \mathbb{R}$. The joint distribution is given by:

$$p(z_1, z_2, z_3, x_1, x_2, x_3) = \frac{1}{Z} \prod_{n=2}^{3} \psi_t(z_n, z_{n-1}) \prod_{n=1}^{3} \psi_e(x_n, z_n),$$

where Z is the partition function, and the transition potential $\psi_t(z_n, z_{n-1})$ and the emission potentials $\psi_e(x_n, z_n)$ are given by:

$\psi_t(z_n, z_{n-1})$	$z_n = F$	$z_n = H$	$z_n = M$
$z_{n-1} = F$	2.0	3.0	5.0
$z_{n-1} = H$	1.0	6.0	3.0
$z_{n-1} = M$	4.5	2.0	2.5

z_1	$\psi_e(x_1, z_1)$
F	1.0
H	8.0
M	1.0

z_2	$\psi_e(x_2, z_2)$
F	7.0
H	1.0
M	2.0

z_3	$\psi_e(x_3, z_3)$
F	2.0
H	3.0
M	5.0

Decode the message that corresponds to the states of the hidden variables that give the maximal probability. Show all your workings clearly.

Question 3 (Lecture 7)

Figure 3.1 shows a Bayesian network of the mixture of Bernoulli distribution. X_n is a binary random variable, i.e. $x_n \in \{0, 1\}$. N is the total number of observations. Z_n is the 1-of-K indicator random variable, $z_{nk} = 1 \Rightarrow z_{n,j\neq k} = 0$ indicates the assignment of the random variable x to the kth Bernoulli density. $z_{nk} \in \{0, 1\}$ and $\sum_k z_{nk} = 1$.

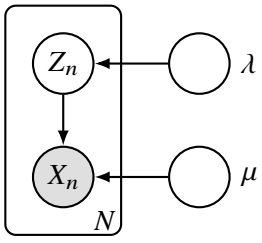


Figure 3.1

Given the expressions for the Bernoulli distribution:

$$p(x \mid \mu) = \prod_{n=1}^{N} \mu^{x_n} (1-\mu)^{(1-x_n)},$$

and marginal distribution of Z_n, which is a categorical distribution specified in terms of the mixing coefficients λ_k:

$$p(z_n) = \prod_{k=1}^{K} \lambda_k^{z_{nk}} = \text{cat}_{z_n}[\lambda], \quad \text{where } 0 \leq \lambda_k \leq 1 \text{ and } \sum_k \lambda_k = 1.$$

(a) Show that the mixture of Bernoulli distribution is given by:

$$p(x \mid \mu, \lambda) = \prod_{n=1}^{N} \sum_{k=1}^{K} \lambda_k \mu_k^{x_n} (1-\mu_k)^{(1-x_n)}.$$

(b) Derive the responsibility $\gamma(z_{nk}) = p(z_{nk} = 1 \mid x)$, and show that the updates for the unknown parameters μ and λ in the maximization step of the EM algorithm are given by:

$$\mu_k = \frac{1}{N_k} \sum_{n=1}^{N} \gamma(z_{nk}) x_n, \qquad \lambda_k = \frac{N_k}{N}, \quad \text{where } N_k = \sum_{n=1}^{N} \gamma(z_{nk}).$$

Show all your workings clearly.

Question 4 (Lecture 12)

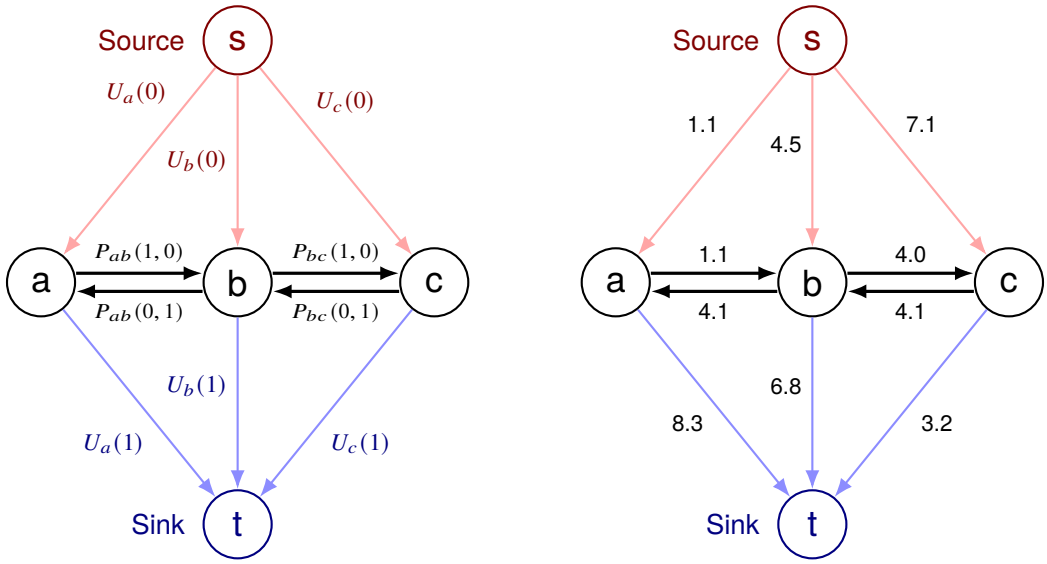


Figure 4.1

(Image source: "Computer Vision: Models, Learning and Inference", Simon Prince)

Compute the **MAP solution** to the three-pixel graph cut problem in Figure 4.1 by

(i) computing the cost of all eight possible solutions explicitly and finding the one with the minimum cost, and

(ii) running the augmenting paths algorithm on this graph by hand and interpreting the minimum cut.

Question 5 (Lecture 9)

Consider the simple 3-node graph shown in Figure 5.1 in which the observed node X is given by a Gaussian distribution $\mathcal{N}(x \mid \mu, \tau^{-1})$ with mean μ and precision τ. Suppose that the marginal distributions over the mean and precision are given by $\mathcal{N}(\mu \mid \mu_0, s_0)$ and $\text{Gam}(\tau \mid a, b)$, where $\text{Gam}(\cdot \mid \cdot, \cdot)$ denotes a gamma distribution. Write down expressions for the conditional distributions $p(\mu \mid x, \tau)$ and $p(\tau \mid x, \mu)$ that would be required in order to apply Gibbs sampling to the posterior distribution $p(\mu, \tau \mid x)$.

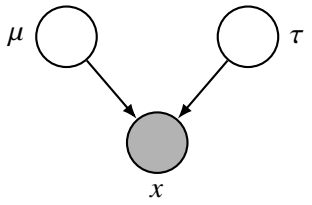


Figure 5.1

Question 6 (Lecture 6)

Figure 6.1 shows a Markov Random Field (MRF) with two random variables X_1 and X_2, where $x_i \in \{0, 1\}$. Furthermore, let $\phi_1(x_1)$ and $\phi_2(x_2)$ denote the unary potentials, and $\psi_{12}(x_1, x_2)$ denotes the pairwise potential. Given the observations over 14 trials as shown in Table 6.1, find the unknown value of $\psi_{12}(x_1 = 0, x_2 = 0)$ in the potential tables shown in Table 6.2. Show all your workings clearly.

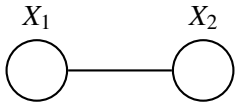


Figure 6.1

Trial Number	Outcomes	
	X_1	X_2
1	0	0
2	1	0
3	1	1
4	1	0
5	0	0
6	0	1
7	1	1
8	0	0
9	1	0
10	1	1
11	0	0
12	0	0
13	1	0
14	1	1

Table 6.1

X_1	X_2	$\psi_{12}(x_1, x_2)$
0	0	$\psi_{12}(x_1 = 0, x_2 = 0)$
0	1	1
1	0	2
1	1	2

X_1	$\phi_1(x_1)$
0	2
1	1

X_2	$\phi_2(x_2)$
0	1
1	2

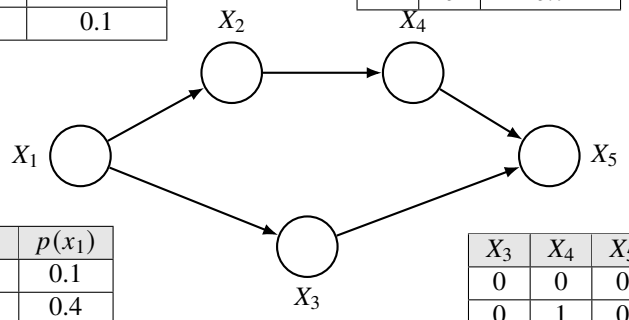
Table 6.2

Question 7 (Lecture 9)

The Bayesian network shown in Figure 7.1 has five random variables X_1, X_2, X_3, X_4, X_5, where $x_i \in \{0, 1, 2\}$ for $i = 1, 2$ and $x_i \in \{0, 1\}$ for $i = 3, 4, 5$.

X_1	X_2	$p(x_2 \mid x_1)$
0	0	0.1
1	0	0.4
2	0	0.8
0	1	0.2
1	1	0.4
2	1	0.1

X_2	X_4	$p(x_4 \mid x_2)$
0	0	0.2
1	0	0.5
2	0	0.7



X_1	$p(x_1)$
0	0.1
1	0.4

X_1	X_3	$p(x_3 \mid x_1)$
0	0	0.3
1	0	0.4
2	0	0.1

X_3	X_4	X_5	$p(x_5 \mid x_3, x_4)$
0	0	0	0.6
0	1	0	0.3
1	0	0	0.1
1	1	0	0.8

Figure 7.1

(a) Given the following numbers drawn from a uniform distribution $u \sim \mathcal{U}(0, 1)$:

$$u = \begin{bmatrix} 0.4387 & 0.4898 & 0.7513 & 0.4984 & 0.2760 \end{bmatrix},$$

generate one set of samples from the joint distribution $p(x_1, x_2, x_3, x_4, x_5)$ using Gibbs sampling. Use $x_1 = 0,\ x_2 = 0,\ x_3 = 0,\ x_4 = 0,\ x_5 = 0$ as the initialization. Show all your workings clearly.

(b) Table 7.1 shows 10 sets of samples drawn from Gibbs sampling. Ignoring the burn-in effect and initialization, find the approximation for the following probabilities using the generated samples:

i. $p(x_2)$

ii. $p(x_3, x_5)$

iii. $p(x_3, x_4 = 1, x_5 = 1)$

iv. $p(x_3 \mid x_2 = 1)$

Sample #	X_1	X_2	X_3	X_4	X_5
0	0	0	0	0	0
1	2	0	1	1	0
2	2	0	0	1	0
3	0	0	0	1	1
4	1	1	1	0	0
5	2	2	1	1	0
6	2	0	1	0	1
7	1	2	0	0	0
8	2	1	0	0	0
9	1	0	1	1	0
10	1	0	1	1	1

Table 7.1

Question 8 (Lecture 8)

a. Figure 8.1 shows a homogeneous hidden Markov Model (HMM) over three timesteps. The latent random variables are Y_1, Y_2, Y_3, where $Y_n \in \{0, 1, 2\}$, and the observed random variables are X_1, X_2, X_3, where $X_n \in \mathbb{R}$.

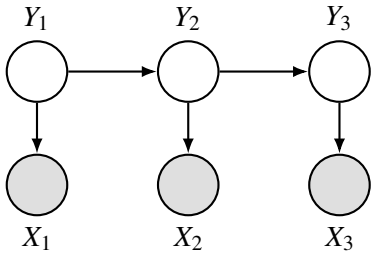


Figure 8.1

The prior probability of the random variable Y_1 is $p(Y_1 \mid \pi) = \prod_k \pi_k^{y_{1k}}$, where $\pi = \{0.2, 0.5, 0.3\}$. Furthermore, the transition probability is given by:

$$p(Y_n \mid Y_{n-1}, A) = \prod_k \prod_j A_{jk}^{y_{n-1,j}\, y_{nk}}, \quad \text{where } A = \begin{bmatrix} 0.2 & \alpha & \beta \\ 0.1 & 0.6 & 0.3 \\ 0.4 & 0.5 & 0.1 \end{bmatrix},$$

and the emission probabilities of the respective observed random variables X_n are shown in Table 8.1.

	$k = 0$	$k = 1$	$k = 2$
X_1	0.3	0.6	0.4
X_2	0.5	0.4	0.4
X_3	0.3	0.8	0.5

Table 8.1

Given that the minimum probability of the joint distribution $p(Y_1, Y_2, Y_3, X_1, X_2, X_3)$ is 0.000216 and occurs at $Y_1 = 0,\ Y_2 = 1,\ Y_3 = 0$, find the unknown values α and β in the transition probability.

b. Figure 8.2 shows an undirected graphical model with six random variables X_1, X_2, X_3, X_4, X_5 and X_6, where $X_i \in \{0, 1, 2\}$. The potential $\psi(X_i, X_j)$ between any pair of nodes X_i and X_j, where $i < j$, is given in Table 8.2. Given $X_1 = 0$, $X_3 = 1$ and $X_5 = 2$, find the states of X_2, X_4 and X_6 that maximize the joint distribution $p(X_1, X_2, X_3, X_4, X_5, X_6)$.

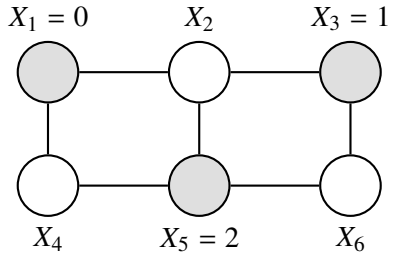


Figure 8.2

X_i	X_j	$\psi(X_i, X_j)$
0	0	1
0	1	5
0	2	7
1	0	2
1	1	4
1	2	8
2	0	3
2	1	6
2	2	9

Table 8.2

Question 9 (Lecture 7)

Figure 9.1 shows a Bayesian network with both binary and continuous state latent random variables, i.e., $Z \in \{0, 1\}$ and $T \in \mathbb{R}$. In addition, $X = 0.5$ is the observed random variable. The maximum log-likelihood of T:

$$\underset{T}{\text{argmax}} \ \log p(T \mid X),$$

can be obtained from the Expectation-Maximization (EM) algorithm. The EM algorithm iterates between the Expectation step that evaluates the expected complete data log-likelihood with respect to $p(Z \mid X, T^{old})$ and the Maximization step that maximizes T over the expected complete data log-likelihood with respect to $p(Z \mid X, T^{old})$. T^{old} is the value of T from the previous iteration of the EM algorithm. $\{\lambda = 0.1,\ w_{a0} = 0.5,\ w_{a1} = 0.5,\ w_{b0} = 0.8,\ w_{b1} = 0.2,\ \tau_a = 1.0,\ \tau_b = 1.2,\ U = 0.6\}$ are known hyperparameters of the following distributions:

$$\begin{aligned}
p(Z) &= \lambda^Z (1-\lambda)^{(1-Z)}, \\
p(X \mid T, Z) &= \mathcal{N}(X \mid w_{a0} + w_{a1} T, \tau_a)^Z \, \mathcal{N}(X \mid w_{b0} + w_{b1} T, \tau_b)^{(1-Z)}, \\
\mathcal{N}(X \mid w_0 + w_1 T, \tau) &= \sqrt{\frac{\tau}{2\pi}} \exp\left\{-0.5\,\tau\,(X - w_0 - w_1 T)^2\right\}, \\
p(T) &= U.
\end{aligned}$$

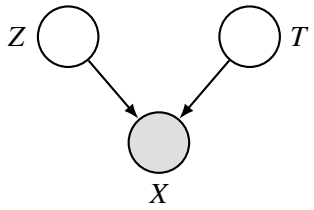


Figure 9.1

a. Derive the expression for the posterior $p(Z \mid X, T^{old})$ from the Bayesian network.

b. Derive the expression for T that maximizes the expected complete data log-likelihood with respect to $p(Z \mid X, T^{old})$.

c. Given the initial value of $T = 2.0$, find the value of T in the next EM iteration.

Question 10 (Lecture 3 and 9)

a. The objective of image denoising is to recover the clean image (noise-free) from a given noisy image. Figure 10.1 shows a Markov Random Field (MRF) to solve a four-pixel binary image denoising problem. The latent random variable $X_i \in \{-1, +1\}$ represents the pixels of the desired clean image, and the observed random variables $Y_i \in \{-1, +1\}$ represent the pixels of the noisy image. We use the Ising model, i.e., $\psi(X_i, X_j) = \exp(JX_iX_j)$ as the edge potentials, where J is the coupling strength of the smoothness prior between neighboring pixels X_i and X_j. The observation model follows a Gaussian distribution:

$$p(Y_i \mid X_i) = \mathcal{N}(Y_i \mid X_i, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-0.5\, \frac{(Y_i - X_i)^2}{\sigma^2}\right\}.$$

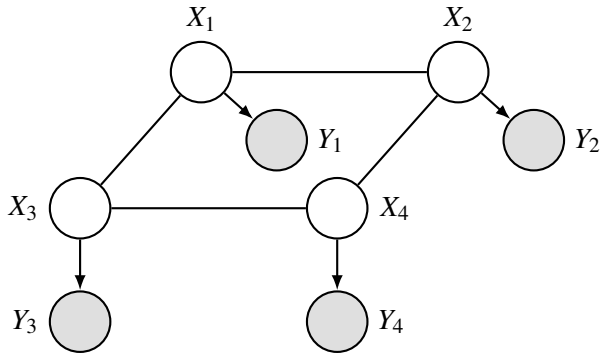


Figure 10.1

Given that we observe $Y_1 = +1,\ Y_2 = -1,\ Y_3 = -1,\ Y_4 = +1$, and the following random numbers are drawn from a uniform distribution $u \sim \mathcal{U}(0, 1)$:

$$u = \begin{bmatrix} 0.6557 & 0.0357 & 0.9340 & 0.8491 \end{bmatrix},$$

generate one set of samples from the joint distribution $p(X, Y)$ using Gibbs sampling. Use $X_1 = -1,\ X_2 = -1,\ X_3 = -1,\ X_4 = -1$ as the initialization and set $J = 0.01$, $\sigma^2 = 1.0$. Show all workings clearly.

b. Draw the Bayesian network and write down the factorized joint probability distribution that encodes all the following conditional independences:

1. $X_4 \perp \{X_1, X_2, X_5\} \mid X_3$
2. $X_5 \perp \{X_1, X_3, X_4\} \mid X_2$
3. $X_3 \perp X_5 \mid \{X_1, X_2\}$
4. $X_1 \perp X_2 \mid \varnothing$
5. $X_1 \perp \{X_2, X_5\} \mid \varnothing$

Question 11 (Lecture 7)

Table 11.1 shows nine observations $\{\mathbf{x}_1, \ldots, \mathbf{x}_9\}$ of 2-dimensional features $[x, y]$, where each observation is generated from an image of 1-out-of-3 handwritten alphabets. We further assume the sampling of each image is fully independent, and the observations given the alphabet follow a bivariate Gaussian distribution (see Equation 1). Figure 11.1 shows a plot of the nine 2-dimensional features in Table 11.1. Given a new observation $\mathbf{x}_{Test} = [14.65,\ 11.00]$, find the probability distribution of the alphabet on its corresponding image. **Explain and show all your workings clearly.**

$\mathbf{x}_n$	$[x, y]$
$\mathbf{x}_1$	[3.83, 14.48]
$\mathbf{x}_2$	[0.31, 2.06]
$\mathbf{x}_3$	[13.62, 8.89]
$\mathbf{x}_4$	[5.74, 1.35]
$\mathbf{x}_5$	[4.02, 15.69]
$\mathbf{x}_6$	[11.82, 9.88]
$\mathbf{x}_7$	[12.39, 10.8]
$\mathbf{x}_8$	[1.64, 15.22]
$\mathbf{x}_9$	[1.84, 0.68]

Table 11.1

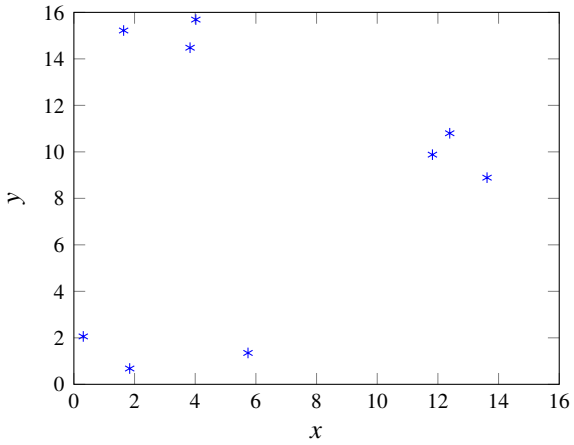


Figure 11.1

Useful Equations:

1. $p(\mathbf{x}) = (2\pi)^{-1} \det(\Sigma)^{-0.5} \exp\left\{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T \Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})\right\}$,
2. $\det \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{bmatrix} = \sigma_{11}\sigma_{22} - \sigma_{12}\sigma_{21}$,
3. $\begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{bmatrix}^{-1} = \dfrac{1}{\sigma_{11}\sigma_{22} - \sigma_{12}\sigma_{21}} \begin{bmatrix} \sigma_{22} & -\sigma_{12} \\ -\sigma_{21} & \sigma_{11} \end{bmatrix}$.

Question 12 (Lecture 10)

Figure 12.1 shows a three time-step Hidden Markov Model (HMM) with binary-state latent $Z_t \in \{0, 1\}$ and observed $X_t \in \{0, 1\}$ random variables. The local conditional and prior probabilities of the HMM are shown in Table 12.1. Using variational inference, find the approximate posterior distribution of $p(Z_1, Z_2, Z_3 \mid X_1, X_2, X_3)$ using the mean-field approximation, i.e. $q(Z_1, Z_2, Z_3) = \prod_{t=1}^{3} q_t(Z_t)$ in one iteration. Assume the initial value of $q_2(Z_2 = 0) = 0.5$, and $X_1 = 0,\ X_2 = 1,\ X_3 = 0$. **Explain and show all your workings clearly.**

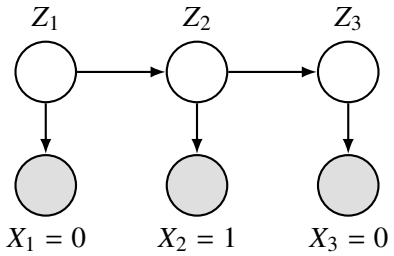


Figure 12.1

z_1	$p(z_1)$
0	0.2
1	0.8

z_1	z_2	$p(z_2 \mid z_1)$
0	0	0.3
0	1	0.7
1	0	0.9
1	1	0.1

z_2	z_3	$p(z_3 \mid z_2)$
0	0	0.4
0	1	0.6
1	0	0.8
1	1	0.2

x_t	z_t	$p(x_t \mid z_t)$
0	0	0.6
0	1	0.7
1	0	0.4
1	1	0.3

Table 12.1

Question 13 (Lecture 10)

Figure 13.1 shows a three-node undirected graphical model, where $X_i \in \mathbb{R}_{\geq 0}$, $\psi(X_1, X_2) = \exp\{-\alpha X_1 X_2\}$, $\psi(X_i) = \exp\{-\beta X_i\}$, and $\alpha = 0.5$ and $\beta = 2.5$ are constants.

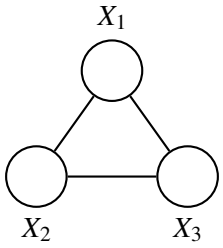


Figure 13.1

a) Using variational inference, find the expressions of the expectation of X_1, X_2, and X_3 under $q(X_1)$, $q(X_2)$, and $q(X_3)$, respectively, where $q(X_1, X_2, X_3) = q(X_1)q(X_2)q(X_3)$ is the mean-field approximation of the posterior distribution $p(X_1, X_2, X_3)$.

b) Taking the initial expected values of X_2 and X_3 under $q(X_2)$ and $q(X_3)$ to be 2.0 and 1.0, respectively, find the mean-field approximation $q(X_1, X_2, X_3)$ after one iteration.

Show all your workings clearly.

Useful Equations:

1. $\displaystyle\int \exp\{kx\}\, dx = \frac{1}{k}\exp\{kx\} + \text{const}$, where $k \neq 0$ is a constant.
2. $\displaystyle\int x\exp\{kx\}\, dx = \left(\frac{kx-1}{k^2}\right)\exp\{kx\}$, where $k \neq 0$ is a constant.

Question 14 (Lecture 9)

Figure 14.1 shows a three-node directed graphical model where $X_i \in \{0, 1\}$. The prior and conditional probabilities are given by the following Bernoulli distributions:

$$p(X_1 \mid \lambda) = \lambda^{X_1}(1-\lambda)^{1-X_1}, \quad \text{where } 0 \leq \lambda \leq 1;$$

$$p(X_2 \mid X_1, \beta) = \beta_{X_1}^{X_2}(1-\beta_{X_1})^{1-X_2}, \quad \text{where } \beta_{X_1} = \begin{cases} \beta_0, & 0 \leq \beta_0 \leq 1, \quad \text{if } X_1 = 0 \\ \beta_1, & 0 \leq \beta_1 \leq 1, \quad \text{otherwise} \end{cases};$$

$$p(X_3 \mid X_1, \gamma) = \gamma_{X_1}^{X_3}(1-\gamma_{X_1})^{1-X_3}, \quad \text{where } \gamma_{X_1} = \begin{cases} \gamma_0, & 0 \leq \gamma_0 \leq 1, \quad \text{if } X_1 = 0 \\ \gamma_1, & 0 \leq \gamma_1 \leq 1, \quad \text{otherwise} \end{cases}.$$

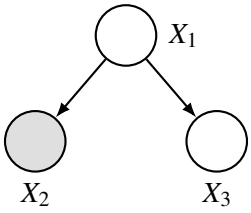


Figure 14.1

a) Given the observation of $X_2 = 1$ and the initial conditions: $\lambda = 0.5$, $\beta_0 = 0.2$, $\beta_1 = 0.6$, $\gamma_0 = 0.3$, and $\gamma_1 = 0.4$. Draw *one set of samples* for X_1 and X_3 using Gibbs sampling. Assume initial samples of $X_1 = 0$, $X_2 = 1$ and $X_3 = 1$, and $u = [0.7655,\ 0.1869]$ are two numbers drawn from a uniform distribution in the range of $[0, 1]$, i.e., $u \sim \text{Uniform}(0, 1)$.

b) Suppose that the subsequent samples drawn from Gibbs sampling are: $\{[X_1 = 1, X_3 = 0], [X_1 = 0, X_3 = 1], [X_1 = 1, X_3 = 1], [X_1 = 0, X_3 = 0]\}$ under the observations of $X_2 = \{0, 1, 1, 0\}$, respectively. Find the updated optimal parameters λ, β_0, β_1, γ_0, and γ_1 with all the samples drawn from Gibbs sampling. Ignore the initial set of samples and the burn-in effect.

Show all your workings clearly.

Useful Equation:

$$\frac{d}{dx}\ln(f(x)) = \frac{1}{f(x)}\frac{df(x)}{dx}.$$

Question 15 (Lecture 8)

Figure 15.1 shows a hidden Markov model with binary latent and observed random variables, i.e. $Z_i \in \{0, 1\}$ and $X_i \in \{0, 1\}$. The prior, transition and emission probabilities are given in Tables 15.1, 15.2 and 15.3, respectively.

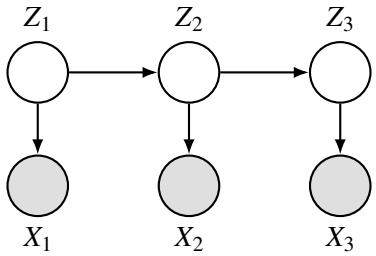


Figure 15.1

Table 15.1

Z_1	$p(Z_1)$
0	0.2
1	0.8

Table 15.2

Z_j	Z_i	$p(Z_j \mid Z_i)$
0	0	0.2
1	0	0.8
0	1	0.7
1	1	0.3

Table 15.3

X_i	Z_i	$p(X_i \mid Z_i)$
0	0	0.4
1	0	0.6
0	1	0.1
1	1	0.9

a) Find the values of a, b and c that give the conditional probability of $p(Z_2 = 0 \mid X_1 = a, X_2 = b, X_3 = c) = 0.579$.

b) Given $X_1 = 0$, $X_2 = 1$, $X_3 = 0$, find Z_1, Z_2 and Z_3 that give the maximum joint probability of $p(X_1, X_2, X_3, Z_1, Z_2, Z_3)$.

Show all your workings clearly.

Question 16 (Lecture 5)

Figure 16.1 shows a loopy graphical model with three binary state latent random variables $X_i \in \{0, 1\}$ and three observed random variables $Y_i \in \{0, 1\}$, where $i = 1, 2, 3$. Tables 16.1 and 16.2 are the likelihood $p(Y_i \mid X_i)$ and the pairwise energy $E(X_i, X_j)$ of the Gibbs distribution, where $\{i, j\} \in \mathcal{E}$ of the graphical model. $\mathcal{E}$ denotes the set of undirected edges in the graphical model.

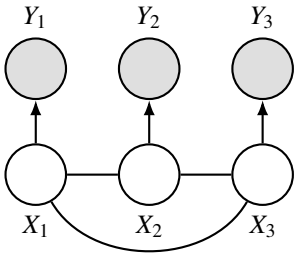


Figure 16.1

X_i	Y_i	$p(Y_i \mid X_i)$
0	0	0.9
1	1	0.9

Table 16.1

X_i	X_j	$E(X_i, X_j)$
0	0	0.01
0	1	1
1	0	1
1	1	0.01

Table 16.2

a) Write the expression for the joint probability of the graphical model, and show that the minimum negative log-posterior over the latent random variables is given by:

$$\underset{X_1,X_2,X_3}{\operatorname{argmin}} \sum_{i=1}^{3} U_i(X_i) + \sum_{\{i,j\}\in\mathcal{E}} S(X_i, X_j),$$

where $U_i(X_i)$ and $S(X_i, X_j)$ are functions of the likelihood and the pairwise potential, respectively.

b) Construct the flow graph that corresponds to the negative log-posterior of the given loopy graphical model.

c) Given that $Y_1 = 1,\ Y_2 = 0,\ Y_3 = 1$, use the flow graph to find the most probable states of the latent random variables X_1, X_2 and X_3.

Show that the most probable states of the latent random variables X_1, X_2 and X_3 can be found alternatively by enumerating all possible states.

Question 17 (Lecture 7)

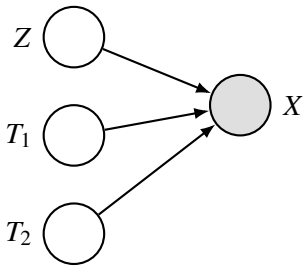


Figure 17.1

Figure 17.1 shows a three-node Bayesian network, where $Z \in \{0, 1\}$ is a binary state random variable that follows the Bernoulli distribution, i.e.:

$$p(Z) = \lambda^Z (1-\lambda)^{(1-Z)}.$$

We further define the conditional probability on the continuous random variable $X \in \mathbb{R}$ to follow a linear-Gaussian distribution, i.e.:

$$p(X \mid T_1, T_2, Z) = \mathcal{N}(X \mid w_{a0} + w_{a1}T_1 + w_{a2}T_2, \tau)^Z \, \mathcal{N}(X \mid w_{b0} + w_{b1}T_1 + w_{b2}T_2, \tau)^{(1-Z)},$$

where

$$\mathcal{N}(X \mid w_0 + w_1T_1 + w_2T_2, \tau) = \sqrt{\frac{\tau}{2\pi}} \exp\left\{-0.5\,\tau\,(X - w_0 - w_1T_1 - w_2T_2)^2\right\}.$$

$T_1 \in \mathbb{R}$ and $T_2 \in \mathbb{R}$ are continuous random variables that follow the uniform distribution, i.e., $p(T_i) = U$ for $i = 1, 2$.

Using the **Expectation-Maximization** algorithm, and taking $\{\lambda, w_{a0}, w_{a1}, w_{a2}, w_{b0}, w_{b1}, w_{b2}, U, \tau\}$ as known hyperparameters, Z as the latent random variable, T_1 and T_2 as the unknown parameters, and X as the observed random variable,

a) Derive the expression for the **posterior distribution** of the latent random variable Z conditioned on the observed random variable X and the old parameters T_1^{old} and T_2^{old}.

b) Derive the expressions for T_1 and T_2 that maximize the expected complete data log-likelihood with respect to the posterior distribution in (a).

Question 18 (Lecture 7)

Figure 18.1 shows the Bayesian network of a mixture-of-Bernoulli distribution. $\mathbf{Z} = [z_1 \ \dots \ z_K]^\top$ is a binary K-dimensional one-hot vector that follows the Categorical distribution parameterized by $\boldsymbol{\pi} = \{\pi_1, \dots, \pi_K\}$, where $\sum_k \pi_k = 1$. X is a binary random variable which follows the k-th Bernoulli distribution indicated by $\mathbf{Z}$. The parameters of the K Bernoulli distributions are $\boldsymbol{\mu} = \{\mu_1, \dots, \mu_K\}$. We are given N *i.i.d.* observations of X.

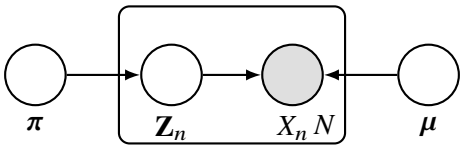


Figure 18.1

(i) Ignoring the prior distributions of the parameters $\boldsymbol{\pi}$ and $\boldsymbol{\mu}$, write the ***complete log-likelihood*** of the given Bayesian network.

(ii) Applying the Expectation Maximization algorithm on the Bayesian network, derive the expression of the ***responsibility function*** in terms of all the random variables and parameters.

(iii) Derive the expressions for the parameters $\boldsymbol{\pi}$ and $\boldsymbol{\mu}$ after the maximization step.

Useful Equations:

1. **Bernoulli distribution**: $p(y \mid \lambda) = \lambda^y (1 - \lambda)^{1-y}$, where y is a binary random variable and the parameter is subjected to $0 \leq \lambda \leq 1$.
2. **Categorical distribution**: $p(\mathbf{y} \mid \boldsymbol{\gamma}) = \prod_k \gamma_k^{y_k}$, where the random variable $\mathbf{y} = [y_1 \ \dots \ y_K]^\top$ is a one-hot vector of K categories, and the parameter is $\boldsymbol{\gamma} = [\gamma_1 \ \dots \ \gamma_K]^\top$ such that $\sum_k \gamma_k = 1$.

Question 19 (Lecture 10)

Given a dataset $\mathcal{D} = \{x_1, \dots, x_N\}$ of observed values of a random variable X, which are assumed to be drawn independently from a univariate Gaussian distribution: $p(x \mid \mu, \tau) = \mathcal{N}(x \mid \mu, \tau^{-1})$ parameterized by the mean μ and precision τ. The prior distributions for μ and τ are given by a univariate Gaussian distribution: $p(\mu \mid \tau) = \mathcal{N}(\mu \mid \mu_0, (\lambda_0 \tau)^{-1})$ and a Gamma distribution: $p(\tau) = \text{Gam}(\tau \mid a_0, b_0)$, respectively. $\{\mu_0, \lambda_0, a_0, b_0\}$ are hyperparameters of the prior distributions.

(i) Using ***variational inference*** and ***mean field approximation***, derive the distributions $q(\mu)$ and $q(\tau)$ that best approximate the conditional distributions $p(\mu \mid \mathcal{D})$ and $p(\tau \mid \mathcal{D})$.

(ii) Derive the expressions for the expected mean μ and precision τ under the approximated distributions $q(\mu)$ and $q(\tau)$.

(iii) Derive the expressions for the mean μ and precision τ that ***maximize the log-likelihood***.

(iv) Comment on the difference between the mean μ and precision τ obtained from variational inference and maximum log-likelihood.

Useful Equations:

1. **Gamma distribution:** $\text{Gam}(\tau \mid a, b) = \dfrac{1}{\Gamma(a)} b^a \tau^{a-1} \exp(-b\tau)$, where $\Gamma(a)$ is the Gamma function;
2. **Gaussian distribution:** $p(x \mid \mu, \tau^{-1}) = \left(\dfrac{\tau}{2\pi}\right)^{1/2} \exp\left\{-\dfrac{\tau}{2}(x-\mu)^2\right\}$;
3. **Completing the square:** $ax^2 + bx + c = 0 \Longrightarrow a(x+d)^2 + e = 0$, where $d = \dfrac{b}{2a}$ and $e = c - \dfrac{b^2}{4a}$.

Question 20 (Lecture 12)

Figure 20.1 shows a hidden Markov model with binary latent random variables Z_1, Z_2, Z_3, Z_4 and binary observed random variables $X_1 = 1$, $X_2 = 0$, $X_3 = 1$, $X_4 = 0$. The unary potentials $U_i(Z_i)$ between $Z_i - X_i$ and pairwise potentials $P_{ij}(Z_i, Z_j)$ between $Z_i - Z_j$ are given in Table 20.1.

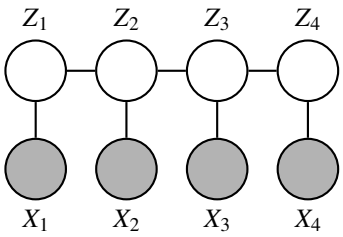


Figure 20.1

Unary Potentials

$U_1(0)$:	20	$U_1(1)$:	5
$U_2(0)$:	5	$U_2(1)$:	20
$U_3(0)$:	20	$U_3(1)$:	5
$U_4(0)$:	5	$U_4(1)$:	20

Binary Potentials

$P_{12}(0,1)$:	10
$P_{23}(0,1)$:	5
$P_{34}(0,1)$:	10
$P_{12}(1,0)$:	5
$P_{23}(1,0)$:	10
$P_{34}(1,0)$:	5

Table 20.1

(i) Using the ***Graph-Cut algorithm***, find the states of the latent random variables Z_1, Z_2, Z_3, Z_4 that maximize the joint probability distribution.

(ii) List all the possible cuts and the respective capacity of each cut.

Question 21 (Lecture 9 and 5)

(i) Figure 21.1 shows the Markov Random Field (MRF) of the binary image denoising problem, where each node represents a binary latent random variable X_t of the clean image. The edge potential is given by the Ising model: $\psi(X_s, X_t) = \exp(J\, X_s X_t)$, where $X_t \in \{-1, +1\}$.

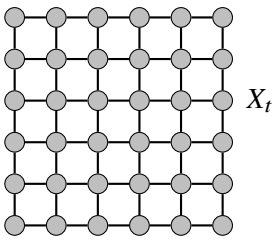


Figure 21.1

Given that each pixel of the observed noisy image $Y_t \in \{-1, +1\}$ is related to the respective latent random variable X_t with a directed edge, i.e., $X_t \rightarrow Y_t$, where the likelihood is a Gaussian distribution $p(Y_t \mid X_t) = \mathcal{N}(Y_t \mid X_t, \sigma^2)$. Both J and σ^2 are fixed hyperparameters.

a) Derive the expression of the full conditional to draw a sample of X_t with ***Gibbs sampling***.

b) Draw a sample of X_t given that all the other latent random variables are $+1$, $Y_t = +1$ and a random number in $[0, 1]$ drawn from a uniform distribution is 0.56. We further take $J = 0.1$ and $\sigma^2 = 0.5$.

(ii) What is the tree-width of the Bayesian network shown in Figure 21.2? Explain your answer clearly.

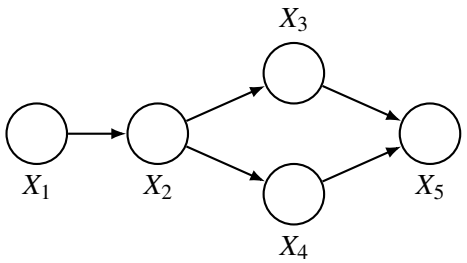


Figure 21.2

Useful Equations:

1. **Gaussian distribution:** $\mathcal{N}(x \mid \mu, \sigma^2) = \left(\frac{1}{2\pi\sigma^2}\right)^{1/2} \exp\left\{-\frac{1}{2\sigma^2}(x-\mu)^2\right\}$;
2. **Sigmoid function**: $\text{Sigm}(x) = \frac{1}{1+\exp\{-x\}}$.

Question 22 (Lecture 7)

We are given two biased coins, Coin A and Coin B, each with an unknown probability of landing heads. In a series of trials, we randomly choose one of the two coins and flip it a single time. However, we ***do not know*** which coin was chosen for each flip.

Let us denote the random variables of choosing each coin as a 1-of-2 representation $Z_k \in \{0, 1\}$, where $Z_{k=0} = 1 \coloneqq$ Coin A and $Z_{k=1} = 1 \coloneqq$ Coin B, and the outcome of a coin flip as $X \in \{0, 1\}$, where $X = 0 \coloneqq$ Head and $X = 1 \coloneqq$ Tail. We further denote the probabilities of choosing Coin B as π, and getting tail for Coin A and Coin B as μ_0 and μ_1, respectively.

a) Given N *i.i.d.* observations of the coin flips, draw the Bayesian network and write the joint probability distribution that represents the coin choosing and flipping process.

b) Using the **EM algorithm**, derive the expressions of μ_0 and μ_1 after the Maximization step.

c) The results of these flips are recorded as a sequence of heads (H) and tails (T): $\{H, T, H\}$. Taking equal probability of choosing either Coin A or B and the initial values of $\mu_0 = 0.6$ and $\mu_1 = 0.5$, perform one iteration of the EM algorithm to update μ_0 and μ_1.

Show all your workings clearly.

Question 23 (Lecture 9 and 10)

a) Let $Y = \{Y_i\}_{i=1}^N$ denote a noisy binary image with N pixels, where each pixel Y_i takes a value of 0 (black) or 1 (white). We want to infer the true underlying image $X = \{X_i\}_{i=1}^N$, where each X_i is also binary. The denoising model assumes the following:

Prior on X: Neighboring pixels in the true image X are likely to have the same value, encouraging spatial smoothness, as represented as a pairwise Markov random field prior:

$$p(X) \propto \exp\Big(\beta \sum_{(i,j)\in\text{neighbors}} X_i X_j\Big),$$

where $\beta > 0$ is a parameter that controls the strength of the smoothness prior and "neighbors" denote all neighbors of each node X_i in the Markov random field.

Observation Model: The probability of observing $Y_i = 1$ given $X_i = 1$ is λ, and the probability of observing $Y_i = 0$ given $X_i = 0$ is also λ, where $0.5 < \lambda \leq 1$ to encourage Y_i in taking the same label as the observation.

Using **variational inference with a mean-field approximation** to approximate the posterior distribution $p(X \mid Y)$:

i. Write the conditional probability distribution for the observation model.

ii. Derive the approximate marginal posterior and the mean value of each pixel i.

b) Figure 23.1 shows the Bayesian network of a **three-component mixture of exponential distribution**. $p(Y \mid \pi) = \prod_k \pi_k^{Y_k}$, where $Y_k \in \{0, 1\}$ is a binary random variable with 1-of-3 representation, and the parameters $\pi = \{\pi_1 = 0.5,\ \pi_2 = 0.3,\ \pi_3 = 0.2\}$. $X \in \mathbb{R}_{\geq 0}$ and $p(X \mid Y, \lambda) = \prod_k (\lambda_k \exp\{-\lambda_k X\})^{Y_k}$, where the parameters $\lambda = \{\lambda_1 = 0.5,\ \lambda_2 = 1.0,\ \lambda_3 = 1.5\}$.

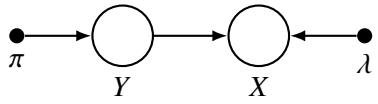


Figure 23.1: Mixture of exponential distribution

i. Given the following 5 pairs of random numbers drawn from a uniform distribution unif$(0, 1)$:

$$\{\{0.76, 0.82\}, \{0.74, 0.69\}, \{0.39, 0.32\}, \{0.66, 0.95\}, \{0.17, 0.03\}\},$$

use **ancestral sampling** to draw 5 sets of samples of (X, Y). We are further given the following useful equation: $\int a \exp\{at\}\, dt = \exp\{at\} + \text{const}$.

ii. Using the drawn samples, find the conditional probability of $p(Y \mid 0 < X \leq 1)$.

Show all your workings clearly.

Question 24 (Lecture 12)

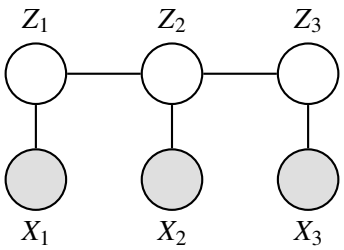


Figure 24.1: Markov Random Field

Table 24.1: Energy Term of Likelihood

X_i	Z_i	$E(Z_i, X_i)$
0	0	2
0	1	6
1	0	6
1	1	2

Table 24.2: Energy Term of Prior

Z_i	Z_j	$E(Z_i, Z_j)$
0	0	1
0	1	4
1	0	4
1	1	1

Figure 24.1 shows a Markov random field (MRF) with three binary latent random variables $Z = \{Z_1, Z_2, Z_3\} \in \{0, 1\}$ and three binary observed random variables $X = \{X_1 = 0, X_2 = 1, X_3 = 0\}$. The joint probability follows the Gibbs distribution:

$$p(Y) = \frac{1}{Z_p} \exp\Big\{ - \sum_{c \in C} E(Y_c) \Big\},$$

where C denotes all cliques in the MRF and Z_p is the partition function. Tables 24.1 and 24.2 give the energies of the likelihood $E(Z_i, X_i)$ and priors $E(Z_i, Z_j)$, where i and j are the neighbors in the MRF.

a) Using the **graph-cut algorithm**, solve for ***all configurations*** of the latent random variables Z for the maximum a posteriori (MAP) probability. Show all details in forming the flow-graph and max-flow min-cut.

b) Using the **max-product algorithm**, solve for ***all configurations*** of the latent random variables Z for the MAP probability.

c) **List exhaustively** the energies that correspond to each configuration of the latent random variables Z. Compare this answer with those in a) and b).

d) State the configuration of Z that gives the ***minimum*** MAP probability and ***explain why*** this configuration leads to the minimum MAP in terms of the prior and likelihood.

Show all your workings clearly.

Question 25 (Lecture 7)

Consider the following hierarchical Bayesian model with a latent class variable $Z_n \in \{1, 2\}$ for each data pair (x_n, y_n), and let $z_n = (z_{n1}, z_{n2})$ be its one-hot encoding with $z_{nk} \in \{0, 1\}$ and $\sum_{k=1}^{2} z_{nk} = 1$:

$$p(z_n \mid \pi) = \prod_{k=1}^{2} \pi_k^{z_{nk}}, \qquad \sum_{k=1}^{2} \pi_k = 1, \quad 0 \le \pi_k \le 1,$$

$$p(x_n \mid z_{nk} = 1) = \mathcal{N}(x_n; \mu_k, \sigma_k^2) = \frac{1}{\sqrt{2\pi\sigma_k^2}} \exp\left(-\frac{(x_n - \mu_k)^2}{2\sigma_k^2} \right),$$

$$p(y_n \mid x_n, z_{nk} = 1) = \mathcal{N}(y_n; a_k x_n + b_k, \tau^2) = \frac{1}{\sqrt{2\pi\tau^2}} \exp\left(-\frac{(y_n - (a_k x_n + b_k))^2}{2\tau^2} \right).$$

All N data points are independent and identically distributed (i.i.d.) according to the parameters: $\theta = \{\pi_k, \mu_k, \sigma_k^2, a_k, b_k, \tau^2\}$. We estimate θ using the Expectation–Maximization (EM) algorithm.

(i) Derive the posterior responsibility $\gamma(z_{nk})$ in the E-step.

(ii) Using the result from the E-step, derive the update equation for each parameter in the M-step.

(iii) Given $N = 3$ data points: $(x_n, y_n) = (1.0, 2.1)$, $(2.0, 3.9)$, $(3.0, 6.1)$ with current parameters: $\pi_1^{(t)} = 0.6$, $\pi_2^{(t)} = 0.4$, $\mu_1^{(t)} = 1$, $\mu_2^{(t)} = 3$, $\sigma_1^{2(t)} = \sigma_2^{2(t)} = 1$, $(a_1^{(t)}, b_1^{(t)}) = (2.0, 0)$, $(a_2^{(t)}, b_2^{(t)}) = (1, 1)$, $\tau^{2(t)} = 0.25$.

Compute the responsibilities $\gamma(z_{nk})$ for each data point numerically (3 decimal places).

(iv) Briefly explain how maximizing the Evidence Lower Bound (ELBO) $\mathcal{L}(q, \theta)$ alternately over q and θ leads to the EM updates derived in (i) and (ii).

Show all your workings clearly.

Useful Equations:

1. $\dfrac{\partial \ln f(x)}{\partial x} = \dfrac{1}{f(x)} \dfrac{\partial f(x)}{\partial x}$.
2. $\dfrac{\partial \exp f(x)}{\partial x} = \dfrac{\partial f(x)}{\partial x} \exp f(x)$.
3. The solution to a linear system of equations: $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$ is:

$$x_1 = \frac{b_1 a_{22} - a_{12} b_2}{a_{11} a_{22} - a_{12} a_{21}}, \qquad x_2 = \frac{a_{11} b_2 - b_1 a_{21}}{a_{11} a_{22} - a_{12} a_{21}}, \qquad \text{given that } a_{11} a_{22} - a_{12} a_{21} \neq 0.$$

Question 26 (Lecture 9)

A renewable-energy company models the daily solar-panel power output Y_t (in kWh) based on two measured covariates: Ambient temperature T_t (°C) and Solar irradiance I_t (W/m^2). The company assumes the following linear-Gaussian model for each day $t = 1, \ldots, 4$: $Y_t = \alpha T_t + \beta I_t + \varepsilon_t$, $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$, where α and β are regression coefficients, and σ^2 represents noise variance. In matrix form, $p(\mathbf{y} \mid X, \theta, \sigma^2) = \mathcal{N}(\mathbf{y} \mid X\theta, \sigma^2 I_N)$, where

$$\mathbf{y} = \begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \\ Y_4 \end{bmatrix}, \qquad X = \begin{bmatrix} T_1 & I_1 \\ T_2 & I_2 \\ T_3 & I_3 \\ T_4 & I_4 \end{bmatrix}, \qquad \theta = \begin{bmatrix} \alpha \\ \beta \end{bmatrix},$$

I_N is an $N \times N$ identity matrix, and N is the number of data points. The priors are: $p(\theta) = \mathcal{N}(0, \Sigma_0)$, where $\Sigma_0 = 10^2 I_2$, and $p(\sigma^2) = \text{Inv-Gamma}(a_0, b_0)$, where $a_0 = b_0 = 2$. The dataset is:

Day t	1	2	3	4
T_t (°C)	28	32	30	29
I_t (W/m^2)	400	600	500	450
Y_t (kWh)	18.5	22.0	19.2	19.7

(i) Assume $\sigma^2 = 1$ is known. The current state of the remaining parameters is $\theta = (0.40,\ 0.008)^\top$, and a proposal $\theta' = (0.60,\ 0.012)^\top$ is drawn from a symmetric Gaussian proposal $q(\theta' \mid \theta) = q(\theta \mid \theta')$.

a) Derive the expression of the Metropolis-Hastings acceptance probability A.

b) Compute numerically the residual sums of squares (RSS): $\|\mathbf{y} - X\theta\|_2^2$ and $\|\mathbf{y} - X\theta'\|_2^2$ using the data table.

c) With $\Sigma_0^{-1} = 0.01 I_2$, compute the log-posterior difference: $\Delta = \log p(\theta' \mid \mathbf{y}) - \log p(\theta \mid \mathbf{y})$, and the acceptance probability A. State whether to accept or reject.

(ii) We now treat σ^2 as unknown while keeping the same Gaussian likelihood $p(\mathbf{y} \mid X, \theta, \sigma^2)$ and prior $p(\sigma^2)$ with $a_0 = b_0 = 2$.

a) We can draw a sample from the conditional posterior of σ^2 given X, θ and $\mathbf{y}$ using Gibbs sampling. Derive this conditional posterior and show that it follows an Inverse-Gamma distribution.

(Hint: Ignore terms independent of σ^2 to simplify the derivation.)

b) Given a sample $u \sim \text{Uniform}(0, 1)$ drawn from a uniform distribution, describe the steps required to draw a sample of σ^2 from its conditional posterior distribution obtained in Part (a).

Show all your workings clearly.

Useful Equations:

1. Inverse-Gamma Distribution Inv-Gamma(a, b):

$$p(x \mid a, b) = \frac{b^a}{\Gamma(a)} x^{-(a+1)} \exp\left(-\frac{b}{x}\right), \quad x > 0, \quad \text{where } \Gamma(2) = 1.$$

2. Multivariate Gaussian distribution $\mathcal{N}(\boldsymbol{\mu}, \Sigma)$:

$$p(\mathbf{x} \mid \boldsymbol{\mu}, \Sigma) = \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})\right), \quad \mathbf{x} \in \mathbb{R}^d.$$

3. Determinant of a 2×2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$: $|A| = ad - bc$.

4. Linear-Gaussian Distribution in matrix form $\mathcal{N}(\mathbf{y} \mid X\theta, \sigma^2 I_N)$:

$$p(\mathbf{y} \mid X, \theta, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{N/2}} \exp\left(-\frac{\|\mathbf{y} - X\theta\|_2^2}{2\sigma^2}\right),$$

where $\|\cdot\|_2^2$ denotes the sum of squared elements.

Question 27 (Lecture 12)

A binary image has N pixels on a 2-D grid. Each pixel i has a latent clean label $X_i \in \{-1, +1\}$ and an observed noisy label $Y_i \in \{-1, +1\}$. The joint distribution is:

$$p(\mathbf{X}, \mathbf{Y}) = \frac{1}{Z_0} \exp\Big(\sum_i L_i(X_i) + \sum_{(i,j)\in E} W_{ij} X_i X_j\Big),$$

where E is the set of neighbor pairs, $W_{ij} > 0$ encourages neighboring pixels to agree, $L_i(X_i)$ is a site-wise log-likelihood term determined by Y_i, and Z_0 is the partition function.

(i) Explain why exact inference of $p(\mathbf{X} \mid \mathbf{Y})$ is computationally intractable on large grids. What structural assumption does the mean-field variational approximation make to enable tractable inference?

(ii) Assume the mean-field factorization $q(\mathbf{X}) = \prod_i q_i(X_i)$ and define $\mu_i = \mathbb{E}_q[X_i]$. Derive the log of the optimized factor $\log q_i^*(X_i)$ up to an additive constant in terms of the neighbor means $\{\mu_j\}_{j\in\text{nbr}(i)}$ and log-likelihood. Clearly indicate which terms depend on X_i.

(iii) Use the sigmoid parameterization:

$$q_i(+1) = \sigma(2a_i), \qquad q_i(-1) = \sigma(-2a_i), \qquad \sigma(z) = \frac{1}{1 + e^{-z}}.$$

Starting from your result in Part (ii):

a) Derive a_i in terms of $L_i(+1)$, $L_i(-1)$, and mean-field influence $m_i = \sum_{j\in\text{nbr}(i)} W_{ij}\mu_j$.

b) Derive a closed-form fixed-point update for the new mean μ_i at pixel i.

(iv) Discuss qualitatively how increasing W_{ij} affects: a) convergence behavior of the mean-field iterations, and b) the smoothness of the final denoised labels.

(v) Consider a 2×2 grid with 4-neighborhoods and uniform $W_{ij} = 0.6$. Let $L_i(X_i) = \beta X_i Y_i$ with $\beta = 1.0$ and

$$\begin{bmatrix} Y_{(0,0)} & Y_{(0,1)} \\ Y_{(1,0)} & Y_{(1,1)} \end{bmatrix} = \begin{bmatrix} +1 & -1 \\ -1 & +1 \end{bmatrix}.$$

Initialize $\mu_i^{(0)} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ and perform two synchronous mean-field updates:

a) Compute $\mu^{(1)}$ and $\mu^{(2)}$ to 3 decimal places.

b) Report MAP labels $\hat{X}_i = \text{sign}(\mu_i^{(2)})$.

Show all your workings clearly.

—End—